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Production scheduling in the context of Industry 4.0: review and trends

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Notwithstanding its disruptive potential, which has been the object of considerable debate, Industry4.0 (I4.0) operationalisation still needs significant study. Specifically, scheduling is a key process that should be explored from this perspective. The purpose of this study is to shed light on the issues regarding scheduling that need to be considered in the new I4.0 framework. To achieve this, a two-stage cascade literature review is performed. The review begins with an analysis regarding the opportunities and challenges brought by I4.0 to the scheduling field, outputting a set of critical scheduling areas (CSA) in which development is essential. The second-stage literature review is performed to understand which steps have been taken so far by previous research in the scheduling field to address those challenges. Thus, a first contribution of this work is to provide insight on the influence and expected changes brought by I4.0 to scheduling, while showcasing relevant research. Another contribution is to identify the most promising future lines of research in this field, in which relevant challenges such as holistic scheduling, or increased flexibility requirements are highlighted. Concurrently, CSA such as decentralised decision-making, and human–robot collaboration display large gaps between current practice and the required technological level of development.

Keywords: scheduling; production planning; Industry 4.0; industry challenges; two-stage literature review

1. Introduction

According to Pinedo (2012), scheduling is a decision-making process in manufacturing and service sectors that deals with allocation of resources (e.g. human, machine, money) to tasks in a certain sequence and over given periods of time. This process is complex, and it aims to optimise operational activities persecuting one or more objectives by leveraging on available production data, which may also include previous scheduling results (i.e. scheduling and rescheduling) (Figure 1). From a mathematical point of view, most of real-world scheduling problems are categorised as NP-hard problems, for which most probably no polynomial-time algorithm exists (Micciancio and Panagiotis 2013). Advances in information and communication technologies (ICT) are increasing the sensitivity of scheduling problems to real-time events, intensifying an already existent trend towards real-time scheduling. This is accompanied by a concurrent increase in the gathering and availability of data on the shop-floor, because of the current trend of using sensors and monitoring instrumentation (Baldea and Harjunkoski 2014; Song and Eldin 2012).

Bringing together many of these advances, the fourth industrial revolution, also known as Industry 4.0 (I4.0), threatens to radically change the traditional production planning and scheduling paradigms (Dolgui et al. 2019). As opposed to previous industrial revolutions, which were essentially triggered by the emergence of a single disruptive technology, I4.0 is instead often defined as a combination of already existing technologies from a wide range of fields. The latter are often related to the broad implementation of ICT in the context of digitalisation of industrial processes and equipment, cyber-physical systems, Internet of things, human–robot collaboration, and real-time big-data processing capabilities (Xu, Xu, and Li 2018; Alcácer and Cruz-Machado 2019). Together, these generate and leverage on the concept of 'smart factories' to comprise the next industrial revolution for manufacturing, characterised by increased flexibility, productivity, efficiency, and sustainability, ultimately ensuring competitiveness in the global market (Kagermann, Wahlster, and Helbig 2013; Rüßmann et al. 2015).

Although a positive and significant contribution of I4.0 to scheduling is expected, the actual transformation that it brings into this field is challenging to analyse and quantify (Harjunkoski 2017; Nahhas et al. 2018). Indeed, not only is the physical organisation of operations being influenced by the new industrial revolution, but the flow of information and its use are also being subjected to significant changes (e.g. information in I4.0 flows both ways between shop-floor and decision-makers, as opposed to traditional scheduling, in which information flows one-directionally towards the shop-floor). All in all, the

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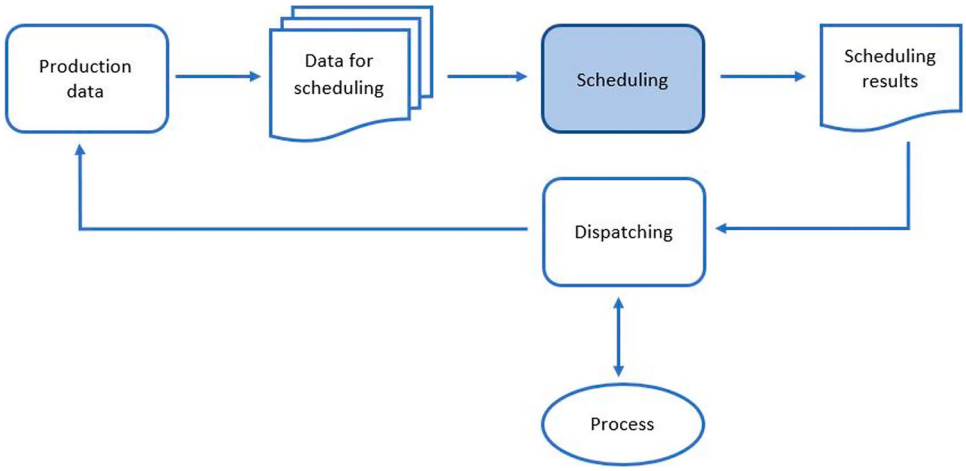


Figure 1. Scheduling process (adapted from Pinedo 2012).

problem of scheduling in the presence of I4.0 is quickly exhausting the capabilities of adaptability of current scheduling practices to the new paradigm, meaning that a fresh perspective regarding the conventional performance of industrial operations may soon be required. Concurrently, from the viewpoint of the scheduler, the paradigm shift yielded by the fourth industrial revolution demonstrates the potential to disrupt the contemporary notion of scheduling itself (Waschneck et al. 2016; Zhang et al. 2017). As such, the main drive behind the present work is the existence of a research gap in scheduling literature related to the influence and potential stemming from the generalised adoption of Industry 4.0. Indeed, it is paramount that scheduling researchers and practitioners become aware of I4.0's impact on scheduling, so as to better direct their future work and scheduling solutions to leverage on its potential. As such, this impact, together with the areas of the scheduling field that are critical to the implementation of I4.0 comprise the focal points of this work. Given the relationship between these two sequential aspects, a two-stage cascade literature review methodology is proposed to achieve the desired goals. This methodology uses the first level literature review to perform an initial exploration of the existing work under the topics related to the impact of I4.0 on scheduling, thus setting up a set of drivers, which are detailed and to which case studies are showcased throughout the second level literature review.

Consequently, the structure of the paper follows the aforementioned two-stage cascade literature review. In the first stage literature review, described in Section 4, a study on the opportunities and challenges brought by I4.0 to scheduling are analysed based on the literature, resulting in a set of *critical scheduling areas* in which development is crucial to undertake the full potential of I4.0 technologies. This stage is supported by a systematic literature review, for which the research methodology is presented in Section 3, and builds upon the discussion of I4.0 and its relation to scheduling presented in Section 2. The second stage of the literature review, corresponding to Section 5, conducts a survey on the current level of development of the scheduling field in each of these *critical scheduling areas*, while performing a comparison with the vision and requirements of I4.0, providing insight on the gap between current research and future needs. Finally, conclusions are drawn in Section 6, including future directions in terms of what still needs to be done in the scheduling field for I4.0 implementation to be possible.

2. Overview of I4.0 and relation to scheduling

The I4.0 concept was devised in 2011 in Germany as part of a strategic manufacturing roadmap to promote the digitalisation of manufacturing. It addresses the current fourth revolution in industries, achieved by means of modern ICT contributions, including pervasiveness, networking, intelligence, autonomy, and scalability (Almada-Lobo 2016; Schmidt et al. 2015). Its introduction is part of a quick evolution of modern technologies and systems in manufacturing, e.g. Computer Integrated Manufacturing, Holonic Manufacturing, Multi-Agent Systems, Autonomous Control and Self-Organised Systems (Leitão, Colombo, and Karnouskos 2016). Originally, I4.0 was one of the projects adopted in the German Federal Ministry of Education and Research's 'Action Plan High-Tech Strategy 2020' (BMBF 2014), which encouraged the establishment of the Platform Industry 4.0 in 2013. In 2015, the platform was expanded with the support of the Ministry for Economic Affairs and Energy. More actors from the private sector, business associations, unions, research organisations and political institutions joined, resulting in a total of over 300 players from 159 organisations to be active in the platform today.

Following the trends of automation and big data in the manufacturing industries, the I4.0 concept was initially supported by three main majors, namely cyber-physical systems (CPS), the internet of things (IoT) and cloud computing (Kagermann, Wahlster, and Helbig 2013; BMBF 2014). I4.0 was envisioned to leverage on an array of technologies to foster 'smart production', allowing CPS, capable of real-time monitoring of physical processes, to communicate in an inter-network of physical devices (IoT), and cooperate with each other to improve production. Having identified its potential as a fourth industrial revolution, several other initiatives have emerged, both from industry and academia. In fact, a noteworthy extension to the I4.0 concept stems from Boston Consulting Group's take on the majors that comprise it (Rüßmann et al. 2015). While also directly featuring ideas like IoT and Cloud-based technologies, this definition complements the majors of I4.0 with concepts such as decentralised and autonomous decision-making (e.g. autonomous robots), big data analysis, simulation (i.e. leveraging on real-time data to model machines, products, and humans, allowing for alternative scenarios testing), horizontal and vertical integration (i.e. integration of not only companies, suppliers, and customers, but also products, shop floor processes, and resources), and cybersecurity (i.e. secure, reliable communications). Other specific technologies are also highlighted in situational contexts, such as additive manufacturing (e.g. 3-D printing) or augmented reality (e.g. virtual reality based systems).

In this framework, as a result of the concepts and technologies that comprise it, the majors of I4.0 can directly support the process of scheduling of activities. Since one of the main drives of I4.0 technologies and methods consists of the collection of data at the edge of physical activities, its integration and systematisation is of the utmost importance in the context of scheduling. The intersection between I4.0 and the scheduling and control of operational activities field is currently the frontier challenge of practitioners and scholars (Ivanov et al. 2016; Klöpper, Pater, and Dangelmaier 2012). However, the definition of I4.0, together with the identification of the majors that govern it, often varies depending on the field of application, as well as on the current level of industrial and technological development of each field. A current example of this can be observed in the manufacturing sector, in which robots are already commonly used to tackle complex assignments. Actually, several manufacturers of robotic equipment already offer interconnected robots that cooperate and interact with one another (ABB 2017; Kuka 2017). As such, in discrete manufacturing industries, development trends already look towards the next steps for implementation of I4.0, such as cloud-connected, autonomous robots carrying out their own optimal scheduling in a decentralised setting (i.e. without central control) (Lang, Fink, and Brandt 2016; Montaqim 2016), or the use of high-end sensors that can not only enable safe work alongside humans, but also autonomously learn from them (Monostori et al. 2016). Contrariwise, in other industries, such as the construction sector, human supervision is generally still regarded as paramount in most processes. For this reason, I4.0 is instead associated with real-time scheduling and control of operations by means of monitoring instrumentation integrated into the construction equipment (Oesterreich and Teuteberg 2016). As such, development trends are directed towards digitalisation of data and processes, by connecting the real-time data gathered from monitoring instrumentation to a virtual model. In turn, this comprises the basis for central control tools, enabling human supervision of construction processes, as well as scenario analysis and optimisation of schedules in the virtual world before translating the results onto their physical counterparts and back into the field (Parente et al. 2018; Song and Eldin 2012).

Bearing this in mind, an updated and generalised conceptualisation of the I4.0 majors for the scheduling field (i.e. independently of the application industry) should be given further consideration. In fact, despite the high relevance of I4.0 in scheduling and the pertinent number of publications on this topic, I4.0 has not been clearly defined (Santos et al. 2017). In contrast to the past three industrial revolutions, for the upcoming fourth revolution no single technology has been identified that triggers Industry 4.0. Instead, it can be described more precisely by a conjunction of many technologies – both existing and new – which now work together (Xu, Xu, and Li 2018). The commonalities and interactions between concepts are depicted in Figure 2, which conveys an updated architecture of the majors behind I4.0 for scheduling, considering recent conceptual and technological developments. The construction of this framework was mainly based on the I4.0 definitions provided by German Federal Ministry of Education and Research (Kagermann, Wahlster, and Helbig 2013; BMBF 2014) and the Boston Consulting Group (Rüßmann et al. 2015), and complemented by the knowledge gathered in the literature review carried out throughout this paper. The main references from the available literature used to interpolate this definition of I4.0 in the context of scheduling are included in the figure in association with the corresponding majors.

Hence, the CPS major was defined as a combination of physical shop-floor agents (i.e. machines, robots, humans), capable of collaborative work (including machine-to-machine and human–robot collaboration), together with the integrated sensors and monitoring instrumentation that gather the status and process data generated during job processing. The 'simulation' major, suggested by Rüßmann et al. (2015), was incorporated into a broader major named adaptive manufacturing, together with the concepts of digitalisation/virtualisation (i.e. the trend to generate virtual models of reality, which can be used for quick and inexpensive scenario analysis together with simulation systems) and decentralised decision-making in order to better reflect the virtual focus of the major. By effectively defining this separation, it becomes possible to

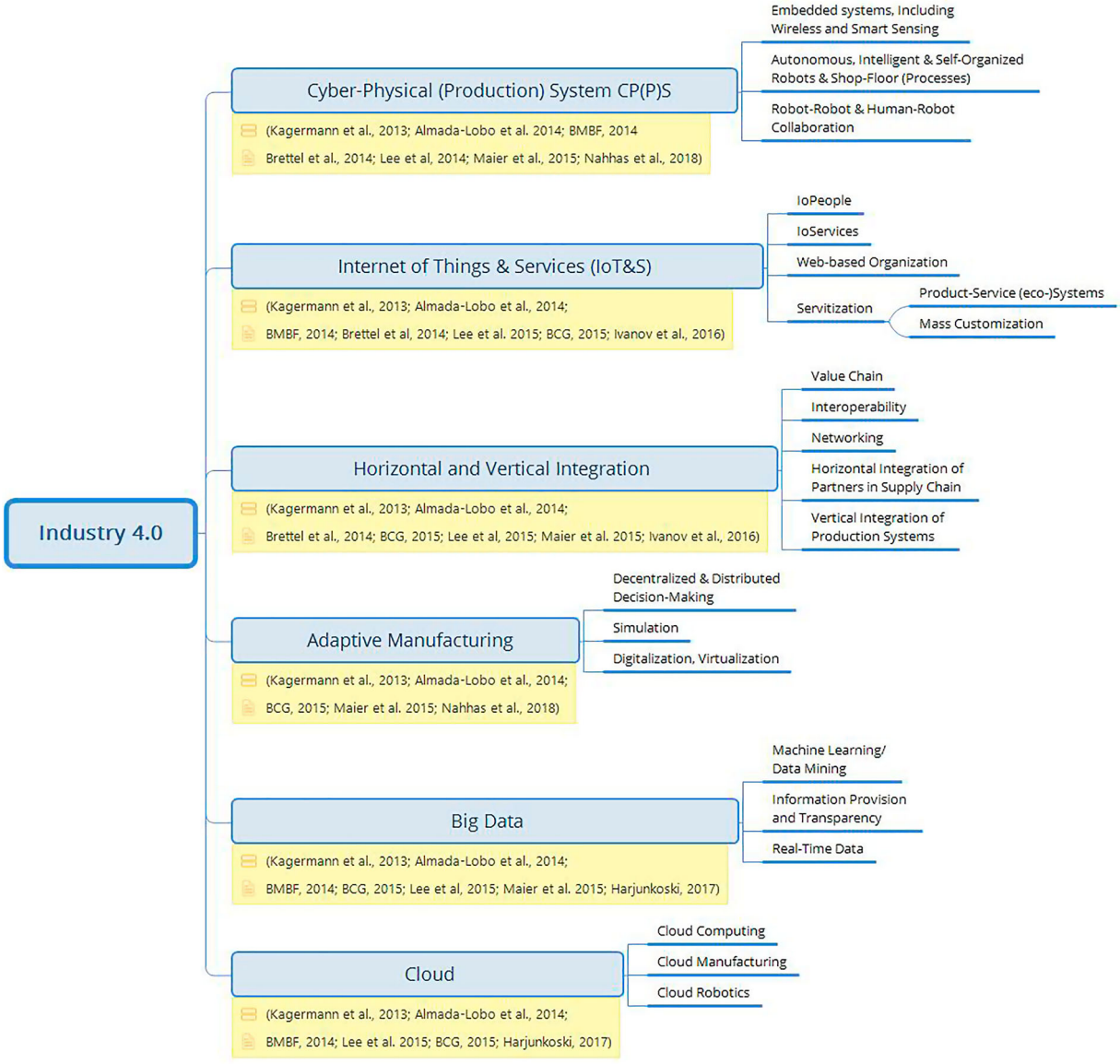


Figure 2. I4.0 concepts and technologies with impact in scheduling.

regard the CPS major as mainly concerned with the physical-world shop-floor agents and associated sensors and monitoring instrumentation, while the Adaptive Manufacturing major addresses the virtual counterpart of these physical resources, modelled by means of simulation systems according to the gathered monitoring data (Nahhas et al. 2018). Even though both majors work closely together, as well as with the remaining majors that comprise I4.0, this separation is crucial to better understand the development trends in different fields. A similar approach was used to interpolate the remaining majors, which are fully consistent with the original definitions, namely the horizontal and vertical integration major, which concerns the integration of competitors and partners in the value chain. The latter major is directly supported by the IoT major, which is defined by the connection to the internet of not only partners, competitors and customers, but also shop-floor agents and decision-makers, effectively enabling majors such as servitisation (i.e. product-service systems, in which customers are in constant contact with manufacturers and suppliers via the internet, placing orders and providing feedback, ultimately facilitating the mass customisation of products which often is used to characterise I4.0) (Ivanov, Sokolov, and Ivanova 2016; Kerin and Pham 2019). For this reason, servitisation was included in the IoT major, and thus renamed as Internet of Things and Services (IoT&S). The Cloud and Big Data majors both interact and support the other majors by providing platforms and technologies that tackle the elevated computational requirements of complex scheduling problems, as well as the analysis of the large amounts of data generated by the other majors in this interconnected environment, respectively (Harjunkoski

2017). Although it could be argued that other concepts mentioned in the literature could also affect scheduling, such as 3-D printing or augmented reality, their application is rather circumstantial, and so these were not considered in this generalised definition of I4.0 in scheduling. This definition of I4.0 and its majors was taken as a reference throughout the remainder of the paper.

3. Research methodology

3.1. Research questions and approach

The goals proposed in this work can be translated into three research questions, namely:

RQ1: How is scheduling affected by I4.0?

RQ2: What has been done so far in the scheduling field that can contribute towards I4.0 implementation?

RQ3: What still needs to be done to allow for a full implementation of I4.0 in scheduling?

The research approach adopted in this paper addresses the discussed issues by following the steps identified in Table 1.

3.2. Literature review methodology

Figure 3 depicts the sequence followed for the two-stage cascade literature review, which reflects the research methodology described in Section 3.1. As previously mentioned, the first level corresponds to a systematic literature review on the opportunities and challenges brought by I4.0 to scheduling, from which a set of critical scheduling areas (CSA) were inferred, comprising the areas in which development is crucial for I4.0 implementation. These CSA also embody the grounds for the analyses in Sections 5 and 6 regarding what has been done in the literature to address the identified challenges, and the issues that still need to be addressed together with future trends, respectively. The analysis in Section 5 includes the assessment of each CSA in terms of its current level of development (LOD) in comparison with the expected development required for deployment in an I4.0 setting.

3.2.1. First-level literature review

According to Fink (2005), a scientific literature review is defined as a fully replicable systematic process. Thus, to carry out the first-level review that supports Section 4, clear boundaries were outlined to delimitate the search:

Table 1. Research methodology and output.

RQ	Section	Description	Resources and methodologies	Outputs
1	4	Identify and analyse the opportunities and challenges brought to scheduling by I4.0	Systematic review/content analysis of scientific databases, such as Scopus, Web of Science, Google Scholar Review/content analysis of white papers and reports from governmental entities or credible/relevant companies	Identify the Critical Scheduling Areas (CSA) that are relevant from an I4.0 point of view
2	5	Assess what has been done so far in the identified critical scheduling areas	Second-level literature review using the critical scheduling areas identified in Step 1 as reference keywords	Insight regarding the directions towards which researchers and industry decision-makers are focusing their efforts
3	6	Assess what still needs to be done to allow for full I4.0 implementation in the scheduling field	The literature review from the previous steps, including future research directions identified and proposed in studied works; Outputs from Step 2 regarding current level of development of critical scheduling areas in comparison to what is expected in I4.0	Identify research gaps between current literature and what is envisioned in I4.0, providing future directions of research Identify relevant technologies to support I4.0 implementation

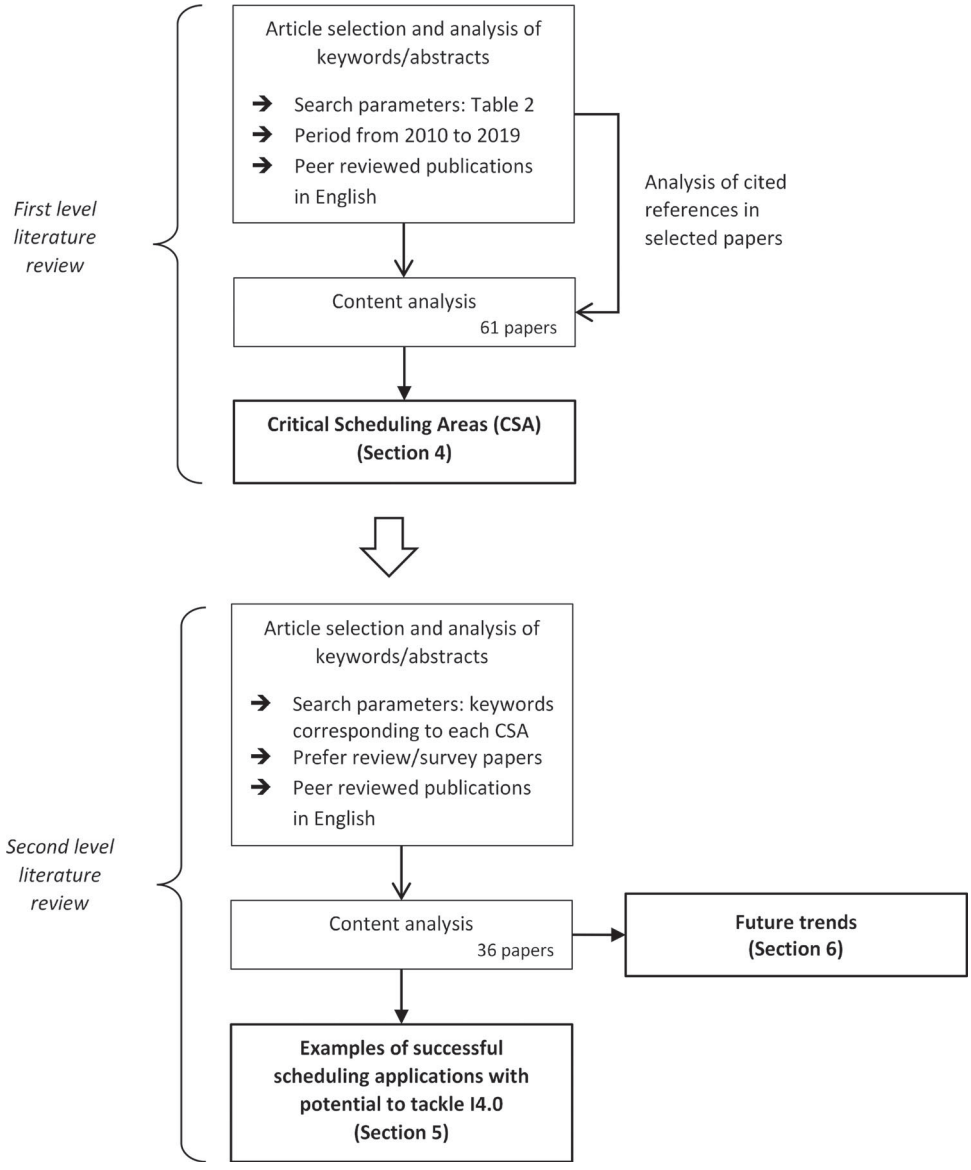


Figure 3. Two-stage cascade literature review.

(1) This analysis included only papers in peer-reviewed scientific journals, conferences and workshops, as well as white papers and reports from governmental entities or credible/relevant companies. Documents with an economic or political focus were excluded.
(2) Despite the existence of a reasonable amount of papers regarding I4.0 in German, many of those authored by the German Federal Government (BMBF), only papers written in English were considered. This consideration is justified by the fact that most of the German language literature is limited to the original German definition of the fourth industrial revolution, while being focused on the application of I4.0 exclusively in the German Industrial context, which is not consistent with the broader purpose of the present paper. Furthermore, the most relevant reports and scientific works originated in Germany were also released as English written documents, meaning that the English literature aptly encompasses the majority of the topics and subjects addressed in this field.
(3) The time period considered for the search ranged from 2010 to 2019. Given that I4.0 is a relatively recent concept, this 10-year period is deemed to be representative of the work developed in the field.

The research was conducted in three phases, which are summarised in Table 2. Ensuing the definition of research goals and scope, the search for related publications was mainly conducted as a structured keyword search (phase 1 in Table 2). Keywords that are susceptible to variations, such as ‘Industry 4.0’ vs. ‘Industrie 4.0’, are represented by the use

Table 2. First-level literature review phases.

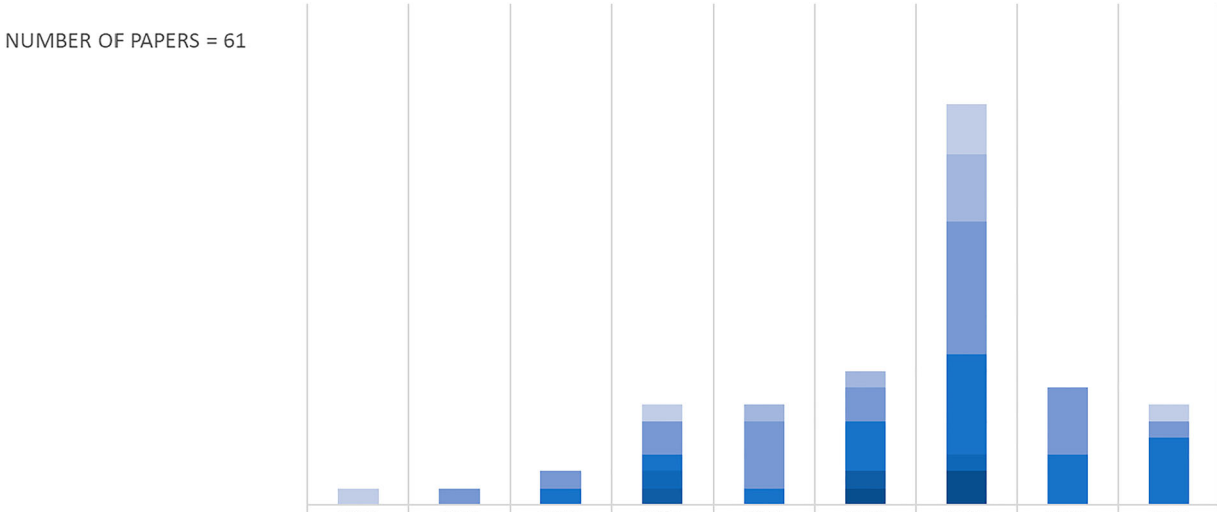
Phase	Nr. of papers	Keywords		Methodology
		Keyword 1	Keyword 2	
1	1303	• Industr* AND 4.0 • 'Fourth industrial revolution' • Technology-related keywords (e.g. 'cyber-physical', 'internet of things')	sched*	Definition of research goals and scope; compilation of information gathered in peer-reviewed journal articles (including reviews), proceedings and technical reports
2	121			Topic and keyword filtering
3	61			Content analysis

of the asterisk (*) symbol in Table 2. In the search, the used keywords are comprised by combinations of words from the first corresponding column of Table 2 (Keyword 1) and the keyword associated with scheduling, shown in the second corresponding column of Table 2 (Keyword 2). The technology-related keywords mentioned in the table correspond to the I4.0 majors previously derived in Section 2, and shown in Figure 2. This data was surveyed in May 2019 through reference scientific databases, namely Elsevier Scopus, Science Direct, Springerlink Journals, IEEE Xplore and Google Scholar. After a first brief topic and keyword check in phase 2, identified articles were included or excluded from the analysis, limiting the large amount of papers from phase 1–105. The final stage consisted of a careful analysis of the content of each paper, selecting only the ones with relevance for the topic at issue. The content analysis was carried out independently by two researchers based on a common referential, the conceptual map of industry 4.0 (c.f. Figure 2). From this literature, both researchers continuously updated the list of opportunities and challenges related to the present topic until all papers were reviewed, at which time they met again for brainstorming on the resulting CSA. Two other researchers were later involved to review the content of the main papers and help to consolidate the inferred CSA. It is important to note that, while reading the papers, cited references were used as a secondary source, but did not yield many additional papers, which can be taken as an indication of the validity of the research. Taking the stated delimitations into account, a total of 55 relevant works were identified, which are enumerated in Annex A.

The content of the papers was assessed by carrying out a descriptive analysis on: (1) how are the publications distributed across the time period? (2) which journals and conferences have published them? (3) which dimensions of scheduling in Industry 4.0 are addressed?

(1) Figure 4 presents the distribution of references according to the year of publication. Its analysis emphasises the novelty and timely relevance of the topic at issue, as inferred from the high amount of publications in the past years (papers since 2017 account for 58% of total of papers reviewed).
(2) Regarding the distribution of references according to publication type and medium (Annex A), even though production engineering and manufacturing are the most predominant fields, journal areas also span through other related subjects, ranging from computer science to communication. This is consistent with the transdisciplinarity of Industry 4.0 applications, which can also be inferred by the identified relevant reports on the subject (Kagermann, Wahlster, and Helbig 2013; BMBF 2014; Rüßmann et al. 2015).
(3) Finally, concerning the majors of scheduling in I4.0 addressed by these publications, each paper was assigned to exactly one category corresponding to its main focus. Figure 4 shows that, while other majors are more homogeneously addressed, the effect of CPS has been under close examination by researchers and industries, which is understandable due to its potential impact. However, several publications, especially reports and review papers, mostly adopt a broad viewpoint of I4.0, covering more than one of its dimensions simultaneously. As such, these were classified as 'Combinations of majors'.

From the relevant review papers, only two of them are specifically dedicated to an analysis of scheduling in the context of I4.0, namely the works by Waschneck et al. (2016) and Zhang et al. (2017). However, the scope of the former is strongly directed and delimited to job shop scheduling in the semiconductor industry, while the latter is focused mostly on the classification of scheduling (e.g. static/dynamic, online/offline, etc.) and its reclassification ensuing the implementation of I4.0. These viewpoints strongly differ from the approach adopted in the current paper, which is focused on firstly studying what impacts are incurred in scheduling by the introduction of I4.0, and secondly on observing how close researchers and the industry are from fully implementing I4.0 in this context. Other works by Harjunkoski (2017) and Nahhas et al. (2018) attempt to adopt a wider perspective. Yet, unlike the present paper that analyses all I4.0 aspects relevant to scheduling as a whole, these works are limited to subsets of aspects related to I4.0, namely the influence of IoT and Horizontal and Vertical Integration in manufacturing, and the opportunities and challenges brought by the combination of CPS and adaptive manufacturing in scheduling, respectively. By addressing all I4.0 aspects and majors in a single paper, it becomes possible



	2010	2012	2013	2014	2015	2016	2017	2018	2019
IoT&S	1			1			3		1
Horizontal and vertical integration					1	1	4		
CPS		1	1	2	4	2	8	4	1
Combination of majors			1	1	1	3	6	3	4
Cloud				1			1		
Big Data				1		1			
Adaptive manufacturing						1	2		

Figure 4. Distribution of reviewed works according to main addressed dimension and year of publication in first-level literature review.

to analyse their correlation and complementarity, which encompass several different fields. This practice was adopted in the current work, thus potentiating cross-fertilisation of concepts and ideas.

3.2.2. Second-level literature review

In what concerns the second-level literature review, supporting Section 5, a broader methodology was adopted, since the focus is on providing a general view rather than discussing specific problems and scheduling methods. Indeed, as previously mentioned, the analysis of Section 5 ensues the delineation of the critical scheduling areas (from the preceding section) and builds upon these by providing a general view of the state of knowledge in each CSA, in relation to the expectations associated with the I4.0 framework. In order to achieve this, a second-level literature review has been carried out by focusing on each CSA individually, seeking methods and applications that indicate potential towards implementation in an I4.0 environment.

Since the purpose of this second-level review is to provide a general insight on the state of development of each CSA within this framework, works with a broader scope were sought after, particularly favouring survey/review papers whenever possible. Despite the preference given to survey/review papers, the search was complemented with other more recent and/or relevant research papers when pertinent. Additionally, while the time period of 2010–2018 may be reasonable for the literature review regarding Industry 4.0, the same restriction was not adopted for the second-level literature review on the critical scheduling areas. Indeed, given the maturity of research in the scheduling field, many works that predate 2010 are still part of the current state of the art. The resulting academic material from the second-level literature review is detailed in Annex B, and further discussed at the beginning of Section 5.

Similarly to the first-level literature review, a descriptive analysis was carried out regarding the publications selected in the second-level review, both in terms of distribution over time (Figure 5) and over publishing journals/conferences. Bearing in mind the purpose of this phase in providing a general view of the state of the art, Figure 5 shows that relevant selected works could be found evenly throughout the years since 2001. These works were complemented with more recent works to provide a better insight on the state of the art in each area, thus increasing the number of analysed publications associated with later years. In what concerns the publishing journals and conferences, Annex B also corroborates the results obtained in the first-level literature review, spanning a wide range of fields, such as engineering, manufacturing, and computer science, which characterise the transdisciplinarity of the topic at issue in this paper.

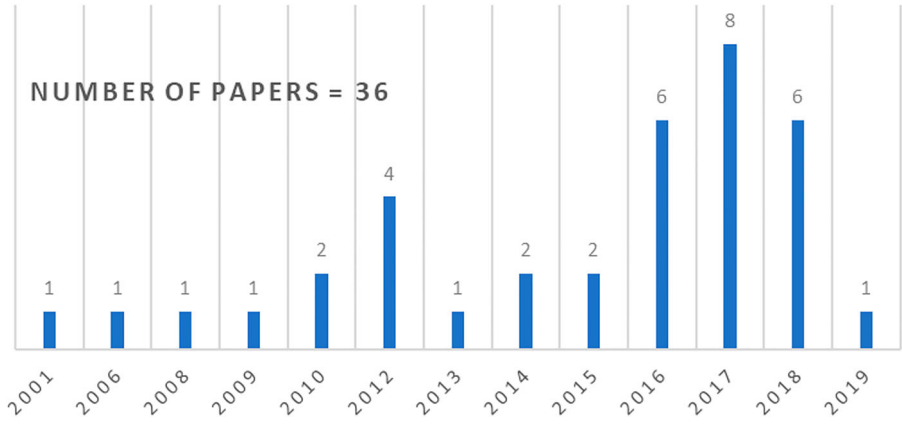


Figure 5. Distribution of reviewed works according to year of publication in second-level literature review.

4. Opportunities and challenges of I4.0 to production scheduling

Bearing in mind the analysis concerning the updated I4.0 majors in Section 1, one can easily infer that some of these majors may bring very different opportunities and challenges to scheduling. While CPS and adaptive manufacturing can arguably represent the most impactful I4.0 majors for scheduling in the operational level (Monostori et al. 2016), scheduling is also heavily influenced by other layers of organisational decision-making. Hence, in order to comprehensively analyse the potential impact of I4.0 on scheduling, the concepts, methods and technologies of I4.0 that affect the other layers of organisation decision-making must be taken into consideration, namely those associated with majors like IoT&S, horizontal and vertical integration, big data, and cloud computing (Kagermann, Wahlster, and Helbig 2013; BMBF 2014; Meissner and Aurich 2019). Table 3 compiles for each I4.0 major the general opportunities relevant to the scheduling field, as well as the most significant challenges associated with each of these opportunities. Additionally, the table features a set of critical research and operational fields (i.e. CSA) related with each challenge. As previously mentioned, these CSA are mentioned in the literature (or otherwise inferred from it by means of content analysis) as high-priority fields from the perspective of the development of scheduling towards Industry 4.0, and are described throughout the text in the ensuing subsections.

Note that, in the scheduling field, big data and cloud technologies (i.e. cloud computing, cloud manufacturing), seem to be mostly regarded as potential solutions to tackle some of the identified challenges, rather than presenting new challenges themselves (Zhong et al. 2017). On the one hand, big data applications provide the capabilities to effectively convert that data into knowledge that can be used to support different problems related to scheduling (e.g. analysing market demand trends, or predicting machine maintenance requirements) (Lee, Kao, and Yang 2014). On the other hand, the tendency of I4.0 for heavier computational requirements or higher global awareness of information highlight the role of cloud computing (Chekired, Khoukhi, and Mouftah 2018; Li et al. 2019). The already important advancements in combining cloud computing with other I4.0 technologies show promise, as is the case of the combination with embedded systems, enabling multi-communication platforms where devices communicate with each other as well as with cloud services (Furuichi and Yamada 2014), or even with big data applications itself, as a way to deal with the sheer amount of data that suddenly becomes available. Simultaneously, cloud manufacturing (i.e. pooling distributed resources to form reconfigurable production lines, enhancing efficiency and agility) is also regarded as a way to respond to the need to quickly adjust to new product portfolios and/or sudden increases in demand (Erol and Sihn 2017; Vincent Wang and Xu 2013).

4.1. Cyber-physical systems (CPS)

4.1.1. Opportunities

Although the majors and concepts of I4.0 are, as previously mentioned, envisioned to work and synergise with each other, taking a closer look at CPS reveals that the implementation of embedded systems, often referred as its most representative feature, is regarded as one of the most impactful aspects of I4.0. Indeed, by leveraging on smart sensors or monitoring technologies for controlling equipment, operations and product flows in real-time, it becomes possible to create a virtual identity of a physical object/resource and connect it to its physical counterpart (Maier, Korbel, and Brem 2015). As a result, an information loop is generated between physical machines and their virtual counterparts that comprise a model of reality, wherein the data and status of the machines is constantly conveyed into the virtual models, while the information stemming from the manipulation of these models can be transmitted back to the machines in real-time (Brettel et al. 2014;

Table 3. Opportunities and challenges in scheduling stemming from the implementation of I4.0

I4.0 major	Section	Opportunities	Challenges	Critical scheduling areas (CSA)
CPS	4.1	Embedded systems, including smart sensors and monitoring. Machine-to-machine / Human–robot collaboration.	To process the large number of events and analyse the large amounts of generated data relevant for scheduling. To account for technological limitations of smart sensors and monitoring technologies in schedules.	Enhanced online, real-time, and reactive scheduling. Holistic scheduling, including higher hierarchical decision levels included in the scheduling process. Machine proactiveness and self-scheduling. Different optimisation trade-offs considering the new interactions. Scheduling under uncertainty, incomplete and missing data.
IoT&S	4.2	Servitisation.	To achieve high schedule adaptability required by mass customisation.	Decentralised and autonomous decision-making. Prominence of the more flexible scheduling paradigms.
Horizontal and vertical integration	4.3	Increased information and transparency.	To integrate the newly available information and concepts into scheduling.	Holistic scheduling, considering more supply chain information and higher hierarchical decision levels included in the scheduling process.
Adaptive manufacturing	4.4	Increased flexibility and effectiveness in complex systems.	To manage, optimise, and schedule in highly complex environments.	Decentralised and autonomous decision-making.

Tantik and Anderl 2016; Rossit, Tohmé, and Frutos 2018). In turn, the machines (or actuators) are responsible for acting according to the processed information. Thus, in these wireless sensor-actuator networks, environmental data is constantly being gathered, interpreted, and acted upon autonomously (Mo et al. 2019).

This has the potential to generate self-aware machines and, subsequently, self-aware machine fleets. Self-awareness is achieved when a machine is capable of recognising itself as part of a whole (i.e. a team or a machine fleet), and is given knowledge regarding what the goal of the team is, as well as what its role is in order to achieve that goal. The implications of self-awareness are clearly evident when considering both machine-to-machine (M2M) collaboration towards a common goal, and human–robot collaboration (HRC). In fact, a system that enables machines to gather information from its peers, human operators/workers, and other surrounding environments can not only enhance the joint productivity of the whole fleet, but also achieve self-awareness of their health by assessing its performance and comparing it its own historical data, as well as those of its peers (Bagheri et al. 2015; Upasani et al. 2017).

4.1.2. Challenges

However, as enticing as the opportunities identified above may be, their application in practice is naturally conditioned by the challenges to their implementation. The use of smart sensors or monitoring technologies comes hand-in-hand with the need for a system that allows the processing of a large number of events and that is capable of efficiently analysing large amounts of data (Maier, Korbel, and Brem 2015; Friedemann et al. 2016). In fact, while the effectiveness of environmental monitoring increases with the amount of sensors in a network, so do the requirements related with efficiency both in terms processing capacity and energy consumption (Ota et al. 2012). Additionally, even though smart sensors and monitoring technologies have been shown to display good performance in controlled conditions, many industries (e.g. construction industry) impose technological challenges to their deployment, since reliable functionality of these systems is in many cases impaired, for instance in the presence of water or large amounts of metal (Brettel et al. 2014; Tang et al. 2015).

From a practical scheduling perspective, the identified opportunities rather call for and compel the development of CSA related to online, real-time and reactive scheduling methodologies, which have been the target of extensive research recently (Hsu and Yang 2016; Zhou, Wang, and Deng 2016; Jing, Xiaoduo, and Hongbo 2017; Lee and Shin 2017; Sousa et al. 2017; Lai et al. 2018; Mourtzis and Vlachou 2018; Zhang et al. 2018), while simultaneously promoting higher-level hierarchical decisions to be included in the online scheduling process from a holistic scheduling viewpoint (Graessler and Poehler 2017; Nahhas et al. 2018). These include a wide array of higher-level decisions, ranging from automatic maintenance scheduling

to reactive adjustment to schedule as a response to unforeseen events (e.g. sudden urgent jobs, or unpredictable delays), and are potentiated by the information stemming not only from CPS applications, but also from other I4.0 concepts, such as IoT&S and horizontal integration. As a consequence, increased flexibility over more organisational dimensions is attained, which also heightens the vertical integration of fleets of machines and production processes. This augmented flexibility and productivity is not just a result of the increased and newly available information: it also stems from the proactiveness of the self-aware machine itself, as it suggests operations to include in schedules based on the data gathered from peers and environment, in view of improving productivity and guaranteeing its own maintenance.

In the more physical oriented features of CPS, the literature indicates that further development is still required in CSA such as machine proactiveness, since there is still a lack of a closely coupled human-machine collaboration (Graessler and Poehler 2017). Indeed, current machines can only passively listen to operators' commands and react, while a smarter machine, the basis for CPS, should be able to actively suggest task arrangements and adjust operational parameters in order to maximise productivity and product quality (Gombolay et al. 2013; Lee, Kao, and Yang 2014). Moreover, known applications of intelligent learning algorithms associated with self-aware and self-maintenance machines show some limitations in terms of handling unprecedented events, preventing them from being widely implemented in the industry (Bagheri et al. 2015).

Altogether, this translates into a different and evolved paradigm in scheduling, pertaining to the fact that the large amount of relevant information, allied with the new interactions between machines and humans, generate a variety of new issues and restrictions that must be taken into consideration during the scheduling process (Schuh et al. 2014; Bochmann et al. 2017). In other words, another relevant CSA is related to the fact that the management of jobs and resources will have to take into account different trade-offs considering these new interactions (Klement, Silva, and Gibaru 2017; Sadik, Taramov, and Urban 2017), for instance by taking the actions and limitations of both collaborative agents (i.e. humans and machines) into consideration in objective functions and/or corresponding restrictions to drive a smooth behaviour of the system. Furthermore, to account for the aforementioned technological limitations of smart sensors and monitoring technologies, scheduling under uncertainty and dealing with missing data will very likely be a reality (Daun et al. 2015; Tang et al. 2015; Fu et al. 2018). Indeed, while most frameworks are adept at handling definitive knowledge, in modern production it is not always possible to make definitive statements about the state of a given system. As even humans have difficulties with uncertain reasoning, scheduling approaches should include rules capable of addressing uncertain knowledge so as to better support decision-making (Kourtis, Kavakli, and Sakellariou 2019). Thus, this aspect has also been pinpointed as a CSA in Table 3.

4.2. Internet of things & services (IoT&S)

4.2.1. Opportunities

Notwithstanding the significant contributions of CPS by itself, it is important to note that the associated interconnectivity verified in the shop floor is enhanced significantly with the inclusion of IoT. Aiming to connect every device to the internet (i.e. including machines, devices, and people), this concept is not limited to supporting and improving the decentralised integration of machines, operators, processes, and decision-makers in the shop floor: it also displays the potential to enable the horizontal integration of products, services and consumers (i.e. servitisation), as well as of suppliers and partners throughout the supply chain through increased availability of information and transparency (Lee, Kao, and Yang 2014; Chen, Tai, and Chen 2017). From the latter applications, servitisation can be defined as a strategic innovation in which an organisation shifts its focus from selling products, to selling integrated products and services, i.e. Product-Service Systems and mass customisation (Martinez et al. 2010; Rüßmann et al. 2015). In practice, the manufacturers' profit sources shift from one-time transactions (i.e. traditional sale of products) to continuous profit and customer service, which can satisfy unmet customer's needs.

This newly achieved connection simultaneously allows customers to customise their own products and provide the company with constant feedback regarding satisfaction. 'Smart products' carry this information and are able to bring the feedback data back to the manufacturing system, which is constantly being improved (Hofmann and Rüsch 2017; Trstenjak and Cosic 2017; Yang and Takakuwa 2017).

4.2.2. Challenges

Nevertheless, to fulfil the potential of using this data, challenges must be overcome. Obviously, the most significant one is related to the fact that the concepts of servitisation and especially mass customisation require highly flexible scheduling capabilities. Hence, it is essential to allow for the schedules' adaptation to different product portfolios, as well as to allow for constant reactive adjustments to accommodate sudden variations in demand or urgent batches. Given the complexity of these requirements, research in CSA related to decentralisation and especially autonomous decision-making are regarded as a crucial means to tackle the corresponding scheduling problems, providing decision-makers with the information that

allows them to carry out their own decisions autonomously (Rüßmann et al. 2015; Trstenjak and Cosic 2017; Yang and Takakuwa 2017).

In such environment, it is only natural that the most flexible scheduling concepts and methods, another CSA identified in Table 3, will gain the uphand over other conventional predictive ones. Indeed, as further discussed in Section 5, concepts such as problem decomposition techniques, or auction-based negotiation methods (Lang, Fink, and Brandt 2016; Pinedo 2012) are expected to gain weight with the implementation of I4.0 (Mourtzis and Doukas 2014; Mrugalska and Wyrwicka 2017; Wu, Tian, and Zhang 2019).

4.3. Horizontal and vertical integration

4.3.1. Opportunities

In what concerns the vertical integration of partners and suppliers in the supply chain, the opportunities for scheduling are directly related to the increase in relevant and instantly available information regarding products, demand, due dates, or resource availability/delays, which become progressively important in the mass customisation era (Erol and Sihn 2017). This can prevent problems related to the traditional central schedule that is rarely up to date and does not take on-site deviations into account. This context typically results in high inventory levels or in carrying out non-value adding activities as components and raw materials are delivered in advance or too late. (Dallasega et al. 2017). Additionally, this large amount of information and data that suddenly becomes available as a result of vertical integration, which includes increased supply chain transparency, also comprises an imperative opportunity for applications of the big data I4.0 major (Lee, Kao, and Yang 2014). This industrial big data, achieved by combining integrated platforms, predictive analysis (e.g. data mining), and visualisation tools, is transformed into useful knowledge that can both support and enhance the scheduling process capability and adaptability (Sokolov and Ivanov 2015), and be used on a more strategic level as a means to gather information on product portfolios or facilitate entrance in new markets (Mourtzis and Doukas 2014).

Concurrently, the horizontal integration addresses the seamless connectivity among all the elements that are included in the product life cycle within an organisation. By closely incorporating activities in marketing, design, engineering, production, and sales, together with those of other companies horizontally, resources within a company (e.g. data, capital, human and machine resources) can be used in a horizontal value network. Ultimately this converts the shop-floor into a dynamic market place in which resources are pooled amongst companies and easily prompted to address different product portfolios, which is translated into dynamic schedules (Chen 2017; Vincent Wang and Xu 2013). Similarly to CPS and IoT&S, both horizontal and vertical integration comprise a way to gather as much as possible the available information that can be used to enhance scheduling. In this case, this information can be targeted at increasing the accuracy and reliability of schedules, as these also account for issues such as delivery delays (i.e. orders, raw materials) or variations in offer and demand.

4.3.2. Challenges

The issues related to this major are no longer so much related with the actual gathering the aforementioned information, but more with how to actually analyse data in order to incorporate it into decision-making from a holistic scheduling perspective, which once again comprises a CSA with significant future repercussions (Ivanov et al. 2016; Pal et al. 2017). Even if, as mentioned before, this comprises an opportunity for big data applications, which can systematise information and turn it into useful knowledge, it also represents a challenge, as these methods are known to have limitations when dealing with massive amounts of data (Lee, Kao, and Yang 2014; Friedemann et al. 2016).

The horizontal integration also brings specific issues into light. Challenges exist in terms of efficiency, intellectual property protection, the establishment of common standards, knowledge sharing, and so forth. The integration of partners and suppliers along the supply chain imply achieving the necessary transparency and trust to enable the open sharing of information between peers. As such, a common network platform with practical protocols and standards is needed in order to jumpstart an effective horizontal integration (Chen 2017). Even though known applications are still not directed at this particular issue, Section 6 includes a discussion on emerging technologies, such as distributed ledger technologies (e.g. blockchain), which could represent a possible solution for this problem (Abeyratne and Monfared 2016).

4.4. Adaptive manufacturing

4.4.1. Opportunities

The adaptive manufacturing systems are responsible for managing all the information originated in CPS, IoT&S, and horizontal/vertical integration in a virtual setting, leveraging on it to impart into scheduling methods the necessary flexibility and increased effectiveness required for tackling the increasingly complex problems of I4.0. While the data from CPS is

more operational-oriented, including on-site processes and resource status, the IoT&S and horizontal and vertical integration majors bring information from other layers of decision-making, such as important data on jobs and restrictions. Both layers are of paramount relevance to scheduling, as the former allows for the development of informed resource management solutions and schedules, and the latter provides the basis for (reactive) adjustments of those schedules as knowledge becomes available (Hajrizi 2016; Nahhas et al. 2018; Wan et al. 2019). Specifically, a fleet of interconnected and self-aware machines allows for task and maintenance plans to be scheduled and optimised at a global level (i.e. for the fleet as a whole). Additionally, given the existence of the aforementioned information loop between production floor and administrative/managerial decision-making, reactive re-scheduling in real-time is enabled as a means of tackling unforeseen events or new information brought in by other majors (e.g. maintenance activities, or urgent production batches) (Tantik and Anderl 2016; Rossit and Tohmé 2018).

4.4.2. Challenges

The foremost identified limitation of the adaptive manufacturing major is mainly conveyed by the difficulties in performing optimisation in such high complexity environments, while simultaneously incorporating the large amount of available data. This is related to the challenges pertaining to enhancing an optimisation system with the capability of effectively decomposing scheduling problems, since the success of most optimisation methods depend heavily on being able to decompose a large problem in sub-problems and still achieve a near-global optimal solution (Waschneck et al. 2016).

This idea leads to one of the CSA preconised by I4.0: the adoption of a decentralised approach for decision-making, both strategically and in the shop-floor (Jiang et al. 2017; Meissner and Aurich 2019). Indeed, given the communication capabilities introduced by I4.0, a decentralised approach would provide the ideal environment for autonomous decision-making to be directly controlled by the shop-floor agents instead of a centralised control unit, increasing production flexibility and agility. The level of decentralisation can be variable depending on the type of application, ranging from partial processing of the gathered data by the agents that harbour the sensors themselves, leaving only a small part of pre-processed data to be shared with a central control unit and/or other agents (also known as Fog or Edge Computing), to full processing and interpretation of all gathered data by the local agents themselves (Mo et al. 2019). This approach enables short decision-making procedures, leveraging on lighter calculation and time requirements, while potentially increasing robustness to deal with failures, since optimal decisions are limited to the extent of an agent's actions (Wan et al. 2019). However, assuming this evolution towards decentralised and autonomous decision-making, e.g. by the use of multi-agent system modelling or dispatching rules (as discussed in Section 5), global optimisation solutions then become an intricate issue (Rossit and Tohmé 2018). If each resource and/or machine is carrying out its own decisions based on local data, an increased effort is then needed to coordinate the single elements as each of them tries to pursue its own goals, which are not necessarily directed towards the global optimisation of the system. Hence, for a decentralised scheduling system to be effective, an alignment of the goals of the individuals is essential to allow them to collectively achieve the goal of the production system (Meissner, Ilsen, and Aurich 2017).

5. An overview of the current level of development of the identified critical scheduling areas

Following the discussion of the opportunities and challenges posed by the different I4.0 majors, Table 4 enumerates the resulting CSA topics according to the sequence of analysis followed throughout this section, as well as the literature that directly addresses them. It is important to note that the discussion regarding different CSA topics is carried out jointly throughout Sections 5.1–5.3, since the corresponding areas are considered to either be closely related or to share important synergies. The rationale for the joint discussion of the identified CSA is as follows:

- Section 5.1 explores the current development of the CSA related to decentralisation and autonomous decision-making. Bearing in mind that decentralisation implies a high level of flexibility, the discussion on this section highlights those developments in the scheduling field capable of tackling decentralised and autonomous decision-making (CSA1) while ensuring the highest scheduling flexibility (CSA2).
- Section 5.2 addresses the integration of the most possible available information into the scheduling process (i.e. holistic scheduling – CSA3). However, since this information is real-world data which is constantly being made available over time, the discussed holistic scheduling techniques focus on those capable of not only real-time scheduling (CSA4), but also capable of dealing with missing or incomplete information (CSA5), which is commonly associated with real-world data.
- Section 5.3 focuses on the assessment of the present level of development of scheduling in the new collaborative environments fostered by I4.0, namely concerning HRC and M2M. Thus, while the analysis regarding the former

Table 4. Critical scheduling areas, as well as associated references and relative level of development.

Critical Scheduling Area (CSA)	Section	References	Relative level of development regarding I4.0 implementation
CSA1 Decentralised and autonomous decision-making	5.1	Adeyeri, Mpofu, and Adenuga Olukorede (2015)* Basnet and Mize (1994) Homberger and Fink (2017) Hsu and Yang (2016) Lang, Fink, and Brandt (2016) Leung, Pinedo, and Wan (2010) Leitão (2009)* Pinedo (2012)* Shen et al. (2006)* Wang, Wan, Zhang, et al. (2016) Wang, Wan, Li, et al. 2016	+
CSA2 Prominence of the more flexible scheduling paradigms			+ +
CSA3 Holistic scheduling	5.2	Baldea and Harjunkoski (2014)* Chen, Yuan, and Chen (2017b)* Kalra and Singh (2015)* Lin et al. (2019) Monch, Fowler, and Mason (2013)* Mouelhi-Chibani and Pierreval (2010) Nguyen, Mei, and Zhang (2017)* Olafsson and Li (2010) Ota et al. (2012) Priore et al. (2001)* Shahzad and Mebarki (2016) Shiue, Lee, and Su (2018) Song and Eldin (2012) Xhafa and Abraham (2008)*	+ +
CSA4 Enhanced online, real-time, and reactive scheduling			+ + +
CSA5 Scheduling under uncertainty, incomplete and missing data			+ + +
CSA6 Different optimisation trade-offs considering the new interactions	5.3	Berenstein and Edan (2012) Bochmann et al. (2017) Bogner et al. (2018) Chen et al. (2014) Graessler and Poehler (2017) Gombolay et al. (2018) Lino Ferreira et al. (2017) Mohan and Bhat (2018) Nikolakis et al. (2018) Ore et al. (2016) Tsarouchi et al. (2017a) Tsarouchi et al. (2017b) Zanchettin et al. (2018)	+
CSA7 Machine proactiveness and self-scheduling			+

*Review/Survey.

encompasses studies that address the integration of HRC interactions into scheduling (CSA6), the latter attends to the concepts related with the proactiveness of machines in the scheduling process (CSA7).

The table also summarises the relative level of development associated with each CSA in the context of I4.0 implementation. The comparison is based on the assessment of the development of the literature for each topic in terms of their readiness for implementation in an I4.0 environment, in accordance with the discussion performed throughout this section.

In order to achieve this comparison, three levels of development were considered, ranging from only few conceptual applications being found in the literature (+) to several applications with the potential to being implemented in practice in an I4.0 framework (+ + +). The intermediate level (+ +) pertains to topics in which some applications were found in the literature, even though these still require some development and/or further testing (e.g. outside of controlled laboratory environments) before real-world implementation can be achieved in this context.

5.1. Decentralised scheduling

As previously discussed, decentralisation and autonomous decision-making (CSA1) comprise a means to tackle the elevated production flexibility and complexity requirements that result from the introduction of I4.0. Although research on decentralisation extends much further from an organisational perspective, such as decentralised manufacturing in autonomous and automated facilities (Wang, Wan, Zhang, et al. 2016; Homberger and Fink 2017), from the shop-floor scheduling point of view, which is the main scope of this paper, this concept implies autonomous agents (i.e. machines, robots, humans) capable of carrying out their own decisions locally, ultimately enhancing agility and optimising production. The discussion in Section 4 revealed that this brings new and complex challenges for scheduling, especially in approaching a globally optimal solution if all shop-floor agents are making their own decisions based on local and 'selfish' criteria. So far, the search for definite solutions is still an ongoing issue in the literature, though some common ground has been established: the more flexible the adopted scheduling paradigms (CSA2), the higher the potential to effectively address decentralised scheduling, as a result of increased agility (Lang, Fink, and Brandt 2016; Wang, Wan, Zhang, et al. 2016; Wang, Wan, Li, et al. 2016). In this context, even though researchers have been dedicating their effort towards studying problems with high flexibility requirements for several years, such as flexible flow shop and flexible job shop problems (Basnet and Mize 1994), a deeper look into this subject and underlying issues is now essential bearing in mind the new perspectives associated with I4.0. For instance, not only can a job be allocated to several machines, or a single machine be able to attend to several jobs simultaneously, but these can also be addressed by more than one resource simultaneously (e.g. multi-processor task scheduling/human–robot collaboration), or alternatively be approached different modes (multi-mode task scheduling), as further discussed in Section 5.3. Given this background, from the already existent scheduling paradigms in the literature, the potential of multi-agent systems to address these issues, particularly allied with auction-based negotiation and decomposition methods, has constantly been highlighted (Shen et al. 2006; Leitão 2009; Adeyeri, Mpofu, and Adenuga Olukorede 2015; Lang, Fink, and Brandt 2016; Wang, Wan, Zhang, et al. 2016).

Whereas the auction-based negotiation methods promote interaction and negotiation between independent agents (i.e. machines, jobs) to generate scheduling solutions, decomposition methods attempt to tackle a problem by decomposing it into simpler subproblems. In what regards the use of former, these generally work by emulating market mechanisms, in which the social interaction between shop-floor agents with individual interests translate into a search for optimal scheduling solutions. Typical examples of this mechanism include representing both jobs and machines as agents, and providing job agents with a budget depending on their characteristics (e.g. deadline, processing requirements). Machine agents, which are characterised by a fixed cost per unit time, can submit bids to job agents in order to maximise their own profit resulting from processing jobs, while job agents look towards minimising cost of being processed. The quality of this type of scheduling procedures is based on the rules used by the various agents to determine bids as well as to accept them, which can become very complex depending on the problem characteristics (Leung, Pinedo, and Wan 2010; Pinedo 2012; Homberger and Fink 2017). Lang, Fink, and Brandt (2016) and Homberger and Fink (2017) reveal that, while there are few applications in this field, it is possible to achieve competitive results in comparison with centralised multi-criteria scheduling procedures in the presence of heterogeneous multi-machine scheduling, especially in cases where agents are competing with each other's self-interests. However, despite the attempts of these authors to output generic solutions, these results stem from specific academic computational experiments (e.g. an extension of the single-mode resource-constrained project scheduling problem in Homberger and Fink 2017), and thus far from ready to tackle the complexity of I4.0, as well as make full use of the sheer amount of information brought by it. Hence, authors underline the need and importance of real-world applications in order to advance knowledge in this field (Shen et al. 2006; Leitão 2009; Lang, Fink, and Brandt 2016; Homberger and Fink 2017), despite the difficulties regarding the required investment to implement these emergent approaches in real-world cases. These difficulties mainly result from the fact that such implementations require not only a new way of thinking and approaching the problems, but also the ability to deal with high complexity of systems due to the high degree of interactions involved both in the virtual and in the physical world.

Relevant examples of decomposition techniques include shifting bottleneck heuristics (i.e. machine-based decomposition), rolling horizon method (i.e. time-based decomposition), or hybrid solutions, which in the past have been implemented in environments that require high flexibility (Shen et al. 2006; Pinedo 2012). Machine-based decomposition methods focus on the generation of schedules for each machine independently. In a decentralised setting, in which each agent is generating

its own schedules, this may facilitate the coordination of agents towards a global optimum (e.g. bearing in mind the potential combination with the previously discussed negotiation mechanisms), mitigating the tendency for selfish action and locally optimal solutions (Shen et al. 2006; Leitão 2009). However, more research is needed to materialise its potential for I4.0 implementation, as it is not clear how much of an investment in computing effort is required in function of the type and quantity of agents (Pinedo 2012). Time-based decomposition methods focus on introducing time intervals of a fixed length and considering in each iteration only jobs that are released during a particular interval. This seems to be more intuitively flexible for the complex environment required in I4.0, as the time frame allows for the reorganisation of each agent according to the volatile demand which, in a mass customisation environment, is expected to be constantly changing. Yet, in both machine- and time-decomposition methods, it is not clear how the quality of the solutions of the subproblems may affect the quality of the overall solution, which may represent a hindrance in an I4.0 setting (Pinedo 2012; Hsu and Yang 2016).

5.2. Holistic real-time scheduling

Bearing in mind the symbiosis and synergy of I4.0 majors, one can argue that decentralisation in the fourth industrial revolution makes particular sense when supported by the increase in information from all sources, transforming manufacturing and scheduling into holistic processes. In what concerns scheduling, this holistic manufacturing (**CSA3**) pertains to making use of all the available information during the schedule generation process, including the data originating from CPS, IoT&S and horizontal and vertical integration majors. Obviously, in this context, this will imply that the data analysed by the agents who perform the scheduling process is susceptible of volatility and constant change in a real-time setting, for instance according to the demand in a mass customisation setting, or to the data gathered from peers in the shop-floor by means of monitoring instrumentation. This means that any holistic scheduling method should be online, real-time, and reactive (CSA4). At the same time, since real-world data is many times fuzzy, incomplete, or missing, the adopted scheduling methods must additionally be able to schedule under uncertainty (CSA5). Thus, considering the available scheduling methods and technologies existent today, those that can tackle holistic scheduling under I4.0 are restricted to a very specific group, namely the ones capable of addressing all these requirements (i.e. associated with CSAs 3, 4, and 5) simultaneously. Despite being applied only in centralised scheduling paradigms, the traditional holistic manufacturing has had extensive research over the years, featuring relevant developments that are susceptible to support an I4.0 framework, such as 'look-ahead' policies (Song and Eldin 2012), metaheuristics (Kalra and Singh 2015; Chen, Yuan, and Chen 2017), and machine learning, both in the form of data mining (Olafsson and Li 2010) and hyper-heuristics (Nguyen, Mei, and Zhang 2017) to generate improved dispatching rules.

In this context, it is important to mention that the traditional dispatching rules (e.g. FIFO, shortest processing time, etc) are neither adaptive, self-learning, nor flexible enough to be effective under I4.0. However, extensions of these, such as 'look-ahead' dispatching rules, may provide the necessary foundation for I4.0's smart factories by attempting to account for future job arrivals (Song and Eldin 2012; Monch, Fowler, and Mason 2013). 'Look-ahead' dispatching rules attempt to take into account the available information on future jobs in the dispatching process allowing them to keep the advantages of traditional dispatching rules, being easy to implement and very fast to process, with the addition that they are able to leverage on the constant stream of data provided by I4.0. However, typical applications show a major drawback related to the fact that rules tend to imply a trade-off between achieving a higher performance in a problem-specific setting (i.e. typically more complex dispatching rules), or settling for a lower performance while facilitating generalisation for other problems/products (i.e. typically simpler dispatching rules). Still, the potential of these extended rules is especially interesting when combined with the other scheduling technologies discussed ahead (i.e. metaheuristics and machine learning), potentially enhancing them to fit both the requirements of holistic manufacturing and real-time scheduling, for instance by means of real-time selection of dispatching rules (Priore et al. 2001; Mouelhi-Chibani and Pierreval 2010), or by actually learning and/or generating new dispatching rules in real-time as new information becomes available (Shahzad and Mebarki 2016; Shiue, Lee, and Su 2018).

Metaheuristics (e.g. evolutionary optimisation) are often cited as a method for dealing with the inaccurate, incomplete, and/or missing data issues inherent to real-world data, allowing scheduling under uncertainty based on artificial intelligence (Xhafa and Abraham 2008; Chen, Yuan, and Chen 2017). However, its advantages extend well beyond those, as metaheuristics have shown to be capable of efficiently outputting good quality scheduling solutions, while being able to integrate additional data related to shop-floor monitoring and control instrumentation in order to perform (near) real-time reactive optimisation (Baldea and Harjunkoski 2014; Kalra and Singh 2015), thus potentially being capable of tackling the CSA of holistic manufacturing (CSA3), real-time and reactive optimisation (CSA4), and scheduling under uncertainty (CSA5). Nevertheless, these solutions still bear the limitations inherent to this type of technology, such as being usually slower than

dispatching rules, and not guaranteeing the optimality of generated solutions. Additionally, solving multi-objective optimisation in a real-time setup can be challenging, even though cloud computing can provide a potential solution to tackle the elevated computational requirements, as previously mentioned.

The third relevant approach identified in this framework is machine learning. Conventional applications are typically based on extracting patterns from past databases (i.e. data mining), and applying the subsequent knowledge to predict future demand and/or scheduling requirements (Olafsson and Li 2010; Lin et al. 2019). Yet, the availability of real-time data has provided the opportunity for new applications in this setting, for instance leveraging on real-time sensor data to predict future events and pre-emptively moving resources (i.e. shop-floor agents) to a certain area before events actually occur (Ota et al. 2012). Furthermore, other machine learning techniques, namely hyper-heuristics (e.g. genetic programming), are capable of not only gathering knowledge on past dispatching rules, but also use it to generate new and improved rules (generally by means of evolutionary optimisation), from which new knowledge may come to light (Nguyen, Mei, and Zhang 2017). Since they are based on a combination of metaheuristics and machine learning techniques, hyper-heuristics can be applied to effectively integrate all three concepts identified in this subsection (dispatching rules, metaheuristics, and machine learning). In a I4.0 context, and similarly to metaheuristics, this can address the critical scheduling areas CSA3, CSA4, and CSA5.

5.3. Human–robot and machine-to-machine collaborative scheduling

As one considers the ultimate concept of smart factory envisioned by I4.0, the idea of self-organising, collaborative robots comes to mind. These new interactions (i.e. human–robot and machine-to-machine collaboration) result in trade-offs that need to be explored and optimised (CSA6). Yet, given the novelty and complexity of the field, from the few relevant applications found in the literature, it was possible to infer that researchers are still dedicating a great deal of effort to understand the requirements for scheduling under these conditions, such as defining human–robot collaboration levels (Berenstein and Edan 2012), and optimising layout for enhancing collaboration (Bochmann et al. 2017; Tsarouchim, Michalos, et al. 2017) or minimising biomechanical load, i.e. considering ergonomic issues (Ore et al. 2016). One can infer that these studies are essential, since aspects such as production process, work station layout, and work load distribution between agents (i.e. machines and humans) radically influence working conditions and productivity, which must be known for optimal scheduling to be possible.

Still, despite the lack of generalised solutions, some optimisation and scheduling attempts can be found in the literature that can be used to build a conceptual optimisation framework even if only for specific problems/industries at this point, especially regarding task allocation in assembly systems (Chen et al. 2014; Tsarouchi, Matthaiakis, et al. 2017; Bogner et al. 2018). A few noteworthy exceptions have recently attempted to tackle the initial steps towards building a framework in the context of scheduling in HRC environments. These can be divided into two distinct paradigms: the first focuses on adapting robot scheduling and actions to human activity, while the second regards humans as part of a robot-dominated system. Most works fall within the first paradigm, attempting to establishing a common model to breakdown collaborative jobs, separating them into specific operations (and associated constraints) to be carried out by humans or robots, ultimately supporting dynamic scheduling and re-scheduling of manufacturing operations (Nikolakis et al. 2018). Also within this paradigm, some works resort to predictive or apprenticeship learning methods to respectively predict and adapt to the activity of humans in collaborative tasks (Zanchettin et al. 2018), or capture the decision-making process carried out by human domain experts in conjunction with optimisation (namely mathematical programming techniques to solve mixed-integer linear programmes via branch-and-bound search) (Gombolay et al. 2018), thus providing a basis for coordinating agents towards the optimal completion of sets of tasks. Concerning the second paradigm, highlight is made to the work by Graessler and Poehler (2017) which features the development of a digital twin to represent employees in production scheduling scenarios. Although the validation of these studies shows promising early results, it has been mainly limited to controlled laboratory settings, using synthetic or real-world data from case studies. Moreover, the proposed scheduling models still lack generalisation capabilities, since more processes must be included and tested so as to allow for the simulation of a complete manufacturing process (Nikolakis et al. 2018); while the predictive approaches adopted by Zanchettin et al. (2018) and Gombolay et al. (2018) are inherently limited by the characteristics of learning algorithms, particularly the quality of the available training dataset resulting, which radically influence the capabilities of the algorithms to learn and generalise information.

The vision in the decentralised production of I4.0 not only includes robots autonomously gathering and analysing the data, but also proactively relaying results and suggestions to other machines and humans (**CSA7**). Indeed, such capabilities would stem from machines achieving a level of self-awareness of themselves and their surroundings, as a consequence of sensorisation, as well as monitoring and control instrumentation (Mohan and Bhat 2018). As a matter of fact, even though each agent acts solely upon the information that is relevant for itself, the event data from other agents and occurrences should still be available for analysis and self-learning. As such, the fact that each agent gathers information from all elements around it enables machine proactiveness, in which the agents will react to the available data by suggesting activities for scheduling

themselves. Until now, research work in this field has focused mainly on proactive self-maintenance, in which a machine assesses its own condition and predicts the need for maintenance, hence suggesting maintenance activity to be scheduled (Lino Ferreira et al. 2017). It is interesting to note that, assuming a fully decentralised HRC environment (e.g. where each agent is engaged in the optimisation its own scheduling by resorting to an auction-based scheduling policy), the fact that a machine would require maintenance, relay it accordingly to peers and humans, and then remove itself from the bid pool would cause the global scheduling process to adjust automatically, as the unfit machine would become a maintenance 'job' waiting for processing, while the remaining agents (including those responsible for maintenance) would keep bidding and processing jobs. For now, this type of interaction is still not possible to achieve given the level of development of this field, which clearly exposes a gap between its current development and what it is expected to achieve (e.g. I4.0 envisions a much higher level of communication capabilities, such as agents not only gathering information, but also relaying knowledge). Yet, the current literature on this subject shows steady development towards these goals.

These conclusions, together with those from previous subsections, are summarised in Table 5, providing an overview of each CSA in terms of its current level of development (LOD), and the aspects that still require more development in order to bridge the gap towards the expected LOD for implementation in an I4.0 environment.

6. Conclusions and future directions

This paper reviewed and analysed the impact of Industry 4.0 (I4.0) on scheduling, ultimately aiming to bring insight on the transformations that scheduling models and methods must go through as the concepts and technologies of Industry 4.0 become more and more adopted by the companies. This was achieved by initially performing a literature review on the opportunities and challenges brought by Industry 4.0 to the production management field. Although there has been little interchange of ideas between Industry 4.0 and the scheduling field, the main trends in both areas are the same, especially regarding the search for increased flexibility and the incorporation of the great amount of available data (i.e. stemming both from the shop-floor and from suppliers/partners/customers). Among the major findings of this study, emphasis is given to the challenges that need to be overcome for the concretisation of I4.0 in scheduling. Indeed, most of the challenges related to the shop-floor, specifically those associated with cyber-physical systems and adaptive manufacturing majors (e.g. managing the large amounts of available information stemming from smart sensors, or optimising in such a highly complex environment), were highlighted as extremely relevant for the topic at hand.

Ensuing this analysis, the original literature review was further directed towards understanding what steps have been taken so far in view of the concretisation of Industry 4.0 in scheduling. This is directly related to and supported by the critical scheduling areas (CSA) defined in the analysis of the previous sections. In this context, Figure 6 conveys a general comparison regarding the level of development in each CSA according to both the maturity of the results found in the analysed literature, and the effort required for the implementation of the corresponding CSA in an I4.0 setting. The figure illustrates the areas that are closer to fundamental research, namely CSA 6 and 7, given the relatively low maturity of results discussed in Section 5, as well as the significant effort that still must be undertaken to bridge the gap between the current level of development of the associated fields and the requirements of I4.0. Conversely, CSA 3, 4 and 5 correspond to the areas closer to practical deployment in real-world conditions. Yet, definitive evidence of robust methods is still required to allow the holistic scheduling concept (CSA3) to be reliably implemented in practice, despite the already significant (though scattered) advances in techniques that can tackle the issue. Finally, regarding CSA 1 and 2, even though some rigorous work has been achieved in terms of flexible scheduling methods, effort still needs to be directed towards finding means to make decentralisation and collaborative scheduling viable in more generalised settings. Concerning the studied technologies, this work established multi-agent systems, metaheuristics, machine learning and hyper-heuristics as the approaches that seem most promising for tackling the challenges related to future Industry 4.0 applications.

To summarise, several challenges and research opportunities still remain which can be translated into future research papers:

- *Decentralised and flexible decision-making:* Although there has been some study together with relevant solutions regarding decentralised and autonomous decision-making by shop-floor agents (Homberger and Fink 2017; Lang, Fink, and Brandt 2016; Wang, Wan, Zhang, et al. 2016), most of the research is either specific to a sector or based on simplified computational experiments. Thus, more research is required to establish decentralisation as definite way to address the complexity of I4.0 in the form of generalised frameworks, especially with focus on the effective use of the additional available information, so as to provide clarity on the actual capabilities of decentralised scheduling for the implementation of CPS and adaptive manufacturing systems (Maier, Korbel, and Brem 2015). Furthermore, there is still some way to go before current approaches are able to incorporate a level of flexibility that complies with the requirements of I4.0, especially regarding the re-configurability of systems to respond to the

Table 5. Current LOD in each CSA and gap to overcome in view of the expected LOD in I4.0

CSA	Current LOD	Gaps to overcome
CSA1 Decentralised and autonomous decision-making	Modelling decentralised scheduling environments is possible (i.e. multi-agent systems), but decisions are limited to 'selfish' and local optima.	Agents must be capable of gathering available data in order to approach globally optimal decisions and schedules in decentralised settings. Tackling the associated complexity and computational requirements is also a priority.
CSA2 Prominence of the more flexible scheduling paradigms	Flexible scheduling methodologies have been the focus of research for many years, yet none has been proven to be able to perform in an I4.0 environment (especially in terms of the flexibility, agility and reactiveness required in decentralised settings) so far.	More research is necessary to establish practical and definitive solutions capable of addressing the requirements related to high flexibility, agility, and reactiveness when dealing with different product portfolios and urgent batches (i.e. tackling servitisation and mass customisation).
CSA3 Holistic scheduling	Combinations of techniques and technologies show potential to perform in an I4.0 environment, though no definitive solutions have been established.	Issues such as dealing with the sheer amount of data, processing data into useful knowledge, and heavy computational requirements still need to be addressed to allow for the integration of all available information into the scheduling process.
CSA4 Enhanced online, real-time, and reactive scheduling	Several well-established alternatives for real-time scheduling already exist, especially regarding the use of dispatching rules in combination with other technologies to generate or adjust them (e.g. metaheuristics, hyper-heuristics). Still, the learning process has been conducted essentially in offline settings.	Schedules (and/or dispatching rules) must be generated or adjusted constantly in real-time as new information becomes available. On the other hand, tackling aspects like responsiveness and computational requirements (e.g. from resorting to metaheuristics or hyper-heuristics) still require more research, though technologies such as cloud computing show promise.
CSA5 Scheduling under uncertainty, incomplete and missing data	Relevant alternatives such as metaheuristics and (predictive) machine-learning techniques show potential in supporting scheduling under uncertainty, while dealing with incomplete and missing data.	Research is needed to provide insight on the practicality of metaheuristics and machine learning in I4.0 across different sectors, since the complexity of the problems is increased in many real-world situations where data might be missing or incomplete.
CSA6 Different optimisation trade-offs considering the new interactions	Current research is still mostly focused only on assessing restrictions, requirements and actual productivity in HRC environments. So far, the few attempts at scheduling solutions and different paradigms lack generalisation potential, and testing has been limited to controlled laboratory settings.	Schedules must be able to account for HRC, optimising and enhancing work for all agents in collaborative environments. Research and testing outside of controlled laboratory settings is essential to develop the maturity of the solutions and subsequent results.
CSA7 Machine proactiveness and self-scheduling	So far, machines are only able to passively heed operator's commands and react.	Machine self-awareness must evolve for them to be able to proactively change or influence generated schedules according to data gathered from sensors and peers.

dynamic, non-linear and often chaotic environments which will characterise most industries. Here, the introduction of learning mechanisms has been underlined as a method to potentially support the identification of re-configuration opportunities, as well as the prediction of disturbances (Leitão 2009). In this context, it is interesting to note that hyper-heuristics have recently been implemented for the real-time improvement/optimisation of the behaviour of decentralised agents in logistics (Van Lon, Branke, and Holvoet 2018).

- *Human–robot collaboration:* While cooperation in the context of M2M has been a topic of note throughout the previous sections, the challenges related to human–robot collaboration have not been the subject of breakthrough solutions. In fact, studies in this field have been addressed at understanding the trade-offs of varying the degree of autonomy of the robotic counterparts of collaborative teams, human situational awareness as a function of robot initiative over decision-making (which may ultimately affect productivity, and thus scheduling tendencies), and mechanisms for incorporating human preference in robot-generated schedules (Gombolay et al. 2013, 2017). Yet,

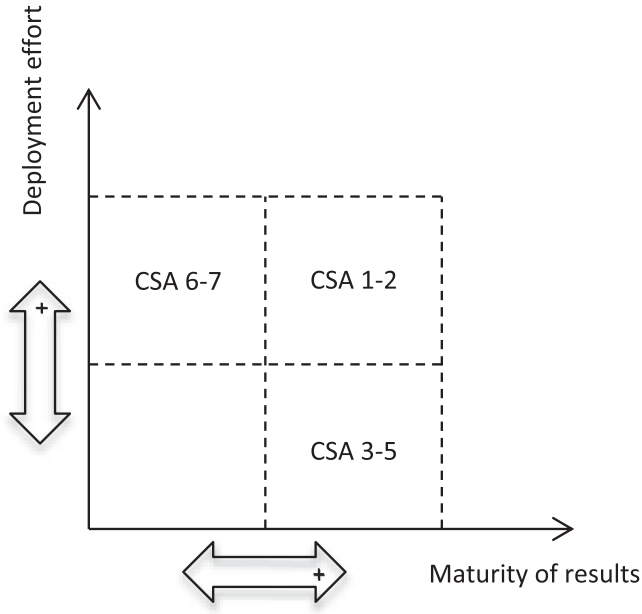


Figure 6. Comparison of the identified CSA according to the maturity of results achieved in the studied literature and the effort still required for implementation in an I4.0 setting.

the analysis in Section 4 still denotes a scarce level of development of the corresponding critical scheduling area in comparison to what is expected in I4.0. Consequently, a greater effort to develop these findings in a generalised setting should be undertaken, so as to overcome this gap and allow for effective and comprehensive models and objective functions capable of taking into account human–robot collaboration environments (e.g. general and flexible simulation-optimisation models of collaborative manufacturing processes). At the same time, it is important to take into consideration the nondeterministic aspect of human nature as a factor in production processes. According to Bogner et al. (2018), studies have mostly approached the behaviour of human workers expecting them to follow the exact work plan and executing the designed tasks at the specified time. However, scenarios in which human workers are given more freedom should also be considered, in which they pick their next task from a set of eligible ones and start it at any time during a specific interval. In practice, this possibility could be supported by the smart sensors and instrumentation provided by CPS in order to provide online input on the current activity of each worker, deploying robot assistance accordingly.

- *Machine proactiveness and self-scheduling:* Future research should also include further exploration regarding machine proactiveness, in the sense that machines are expected to be capable of suggesting changes or supplements to schedules. Studies so far have been mostly focused on proactive self-maintenance scheduling (Lee, Kao, and Yang 2014; Upasani et al. 2017). Yet, as mentioned in Sections 4 and 5, Industry 4.0 preconises an even greater level of machine proactiveness, which stems from their interaction and autonomous decision-making capabilities, and includes actual reactive scheduling of some tasks without human intervention, allowing machines to adjust to the data gathered from themselves, their peers, and the surrounding environment according to their own requirements, as well as the global goals of the production team (Lang, Fink, and Brandt 2016; Mohan and Bhat 2018; Lino Ferreira et al. 2017).
- *Exploration of new technologies:* Naturally, the constant emergence of new technologies creates the potential to address challenges that have not yet been explored in the literature. Distributed ledger technologies, such as the blockchain (Nakamoto 2008), have recently been prompting a lot of interest and attention from both academics and industry. A blockchain is a cryptographically verified linked-list, in the form of a distributed database shared via a peer-to-peer network, that maintains a continuously growing list of ordered records called blocks. The fact that the data added to the blocks is immutable, verified by peers, transparent (i.e. each party can be allowed to see all transactions since the original block), and managed autonomously means that blockchains are highly efficient in recording transactions between two parties in a permanent and safe way (Iansiti and Lakhani 2017). In the context of IoT, other distributed ledger technologies, such as directed acyclic graphs (Popov 2018) promise to be even more scalable and efficient. These technologies synergise with concepts from I4.0's such as horizontal and vertical

integration of processes, products, and services, as well as partners, suppliers, and customers (Abeyratne and Monfared 2016; Weber et al. 2016; Korpela, Hallikas, and Dahlberg 2017; Saberi et al. 2018). Its decentralisation and transparency can not only address issues related to the volatility of markets by improving connectivity and trust, but also provide a secure platform for sharing and retrieving data which can be paramount for augmenting aspects related to holistic real-time manufacturing. A simple and noteworthy example lies in the fact that changes in demand or due dates can be considered automatically and in real-time in the production process (Xu, Xu, and Li 2018).

Finally, it is important to note that the concept of I4.0 and its associated majors are in constant evolution, and, as such, so are their subsequent contributions to scheduling. Thus, it is plausible that additional I4.0 majors other than those addressed in this work can be considered influencing factors to the scheduling field in the near future, which will further open potential research directions and application possibilities.

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Appendix 1. First-level literature review publications

Authors	Title	Published	Year	Type	Main addressed major
Abeyratne and Monfared	Blockchain Ready Manufacturing Supply Chain Using Distributed Ledger.	International Journal of Research in Engineering and Technology	2016	Journal	Horizontal and vertical integration
Bagheri et al.	Cyber-Physical Systems Architecture for Self-Aware Machines in Industry 4.0 Environment.	IFAC-PapersOnLine	2015	Conference	CPS
Bochmann et al.	Human-Robot Collaboration in Decentralized Manufacturing Systems: An Approach for Simulation-Based Evaluation of Future Intelligent Production.	Procedia CIRP	2017	Conference	CPS
Brettel et al.	How Virtualization, Decentrazliation and Network Building Change the Manufacturing Landscape: An Industry 4.0 Perspective.	International Journal of Mechanical, Aerospace, Industrial, Mechatronic and Manufacturing Engineering	2014	Journal	CPS
Chekired et al.	Industrial IoT Data Scheduling Based on Hierarchical Fog Computing: A Key for Enabling Smart Factory	IEEE Transactions on Industrial Informatics	2018	Journal	Combination of majors
Chen	Integrated and Intelligent Manufacturing: Perspectives and Enablers.	Engineering	2017	Journal	Horizontal and vertical integration
Chen et al.	Application of Programmable Logic Controller to Build-up an Intelligent Industry 4.0 Platform.	Procedia CIRP	2017	Conference	IoT&S
Dallasega et al.	Simulation Based Validation of Supply Chain Effects through ICT Enabled Real-Time-Capability in ETO Production Planning.	Procedia Manufacturing	2017	Conference	Horizontal and vertical integration
Daun et al.	Collaborating Multiple System Instances of Smart Cyber-physical Systems: A Problem Situation, Solution Idea, and Remaining Research Challenges	SESCPS ‘15 Proceedings of the 2015 IEEE/ACM 1st International Workshop on Software Engineering for Smart Cyber-Physical Systems	2015	Conference	CPS
Erol and Sihn	Intelligent Production Planning and Control in the Cloud – towards a Scalable Software Architecture.	Procedia CIRP	2017	Conference	Cloud
Federal Ministry of Education and Research (BMBF)	The New High-Tech Strategy Innovations for Germany.		2014	Report	Combination of majors
Friedemann et al.	A Framework for Information-Driven Manufacturing.	Procedia CIRP	2016	Conference	Big Data
Fu et al.	Two-objective stochastic flow-shop scheduling with deteriorating and learning effect in Industry 4.0-based manufacturing system	Applied Soft Computing	2018	Journal	CPS

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Authors	Title	Published	Year	Type	Main addressed major
Furuichi and Yamada.	Next Generation of Embedded System on Cloud Computing.	Procedia Computer Science	2014	Conference	Cloud
Gombolay et al.	Towards Successful Coordination of Human and Robotic Work Using Automated Scheduling Tools: An Initial Pilot Study.	Robotics: Science and Systems, Human-Robot Collaboration Workshop	2013	Workshop	CPS
Graessler and Poehler	Integration of a digital twin as human representation in a scheduling procedure of a cyber-physical production system	2017 IEEE International Conference on Industrial Engineering and Engineering Management (IEEM)	2017	Conference	CPS
Hajrizi	Smart Solution for Smart Factory.	IFAC-PapersOnLine	2016	Conference	Adaptive manufacturing
Harjunkoski	Future of control and operations in the era of industrial internet of things	Computer Aided Chemical Engineering	2017	Journal	Combination of majors
Hofmann and Rüsch	Industry 4.0 and the Current Status as Well as Future Prospects on Logistics.	Computers in Industry	2017	Journal	Combination of majors
Hsu et al.	Real-Time Near-Optimal Scheduling With Rolling Horizon for Automatic Manufacturing Cell	IEEE Access	2016	Journal	Combination of majors
Ivanov et al.	Schedule coordination in cyber-physical supply networks Industry 4.0	IFAC-PapersOnLine	2016	Conference	Combination of majors
Jiang et al.	Distributed Dynamic Scheduling for Cyber-Physical Production Systems Based on a Multi-Agent System	IEEE Access	2017	Journal	Combination of majors
Jing et al.	An improved real-time task preemptive scheduling in cyber-physical systems	2017 29th Chinese Control And Decision Conference (CCDC)	2017	Conference	CPS
Kagermann et al.	Recommendations for Implementing the Strategic Initiative INDUSTRIE 4.0.		2013	Report	Combination of majors
Klement et al.	A Generic Decision Support Tool to Planning and Assignment Problems: Industrial Application & Industry 4.0.	Procedia Manufacturing	2017	Conference	CPS
Kourtis et al.	A Rule-Based Approach Founded on Description Logics for Industry 4.0 Smart Factories	IEEE Transactions on Industrial Informatics	2019	Journal	Combination of majors
Lai et al.	Task Scheduling for Cloud Based Cyber-Physical Systems	2018 IEEE SmartWorld, Ubiquitous Intelligence & Computing, Advanced & Trusted Computing, Scalable Computing & Communications, Cloud & Big Data Computing, Internet of People and Smart City Innovation	2018	Conference	CPS

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Authors	Title	Published	Year	Type	Main addressed major
Lee and Shin	Development and use of a new task model for cyber-physical systems: A real-time scheduling perspective	Journal of Systems and Software	2017	Journal	CPS
Lee et al.	Service Innovation and Smart Analytics for Industry 4.0 and Big Data Environment.	Procedia CIRP	2014	Conference	Big Data
Li et al.	A Hybrid Computing Solution and Resource Scheduling Strategy for Edge Computing in Smart Manufacturing	IEEE Transactions on Industrial Informatics	2019	Journal	Combination of majors
Maier et al.	Innovation in Supply Chains – Solving the Agency Dilemma in Supply Networks by Using Industry 4.0 Technologies.	International Journal of Communication Networks and Distributed Systems archive	2015	Journal	CPS
Martinez et al.	Challenges in Transforming Manufacturing Organisations into Product-service Providers.	Journal of Manufacturing Technology Management	2010	Journal	IoT&S
Meissner and Aurich	Implications of Cyber-Physical Production Systems on Integrated Process Planning and Scheduling	Procedia Manufacturing	2019	Conference	CPS
Meissner et al.	Analysis of Control Architectures in the Context of Industry 4.0.	Procedia CIRP	2017	Conference	Adaptive manufacturing
Mo et al.	Driven Joint Mobile Actuators Scheduling and Control in Cyber-Physical Systems	IEEE Transactions on Industrial Informatics (Early Access)	2019	Journal	Combination of majors
Monostori et al.	Cyber-Physical Systems in Manufacturing.	CIRP Annals	2017	Conference	CPS
Mourtzis and Doukas	Design and Planning of Manufacturing Networks for Mass Customisation and Personalisation: Challenges and Outlook.	Procedia CIRP	2014	Conference	IoT&S
Mourtzis and Vlachou	A cloud-based cyber-physical system for adaptive shop-floor scheduling and condition-based maintenance	Journal of Manufacturing Systems	2018	Journal	CPS
Mrugalska and Wyrwicka	Towards Lean Production in Industry 4.0.	Procedia Engineering	2017	Conference	IoT&S
Nahhas et al.	Toward Adaptive Manufacturing: Scheduling Problems in the Context of Industry 4.0	2018 Sixth International Conference on Enterprise Systems (ES)	2018	Conference	Combination of majors
Ota et al.	ORACLE: Mobility control in wireless sensor and actor networks	Computer Communications	2012	Journal	CPS
Pal et al.	Model-based maintenance scheduling in flexible modular automation systems	2017 22nd IEEE International Conference on Emerging Technologies and Factory Automation (ETFA)	2017	Conference	Horizontal and vertical integration
Rossit and Tohmé	Scheduling Research Contributions to Smart Manufacturing	Manufacturing Letters	2018	Journal	Adaptive manufacturing
Rossit et al.	Industry 4.0: Smart Scheduling	International Journal of Production Research	2018	Journal	CPS

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Authors	Title	Published	Year	Type	Main addressed major
Sadik et al.	Optimization of tasks scheduling in cooperative robotics manufacturing via johnson's algorithm case-study: One collaborative robot in cooperation with two workers	2017 IEEE Conference on Systems, Process and Control (ICSPC)	2017	Conference	CPS
Schuh et al.	Short-Term Cyber-Physical Production Management.	Procedia CIRP	2014	Conference	CPS
Sokolov and Ivanov	Integrated Scheduling of Material Flows and Information Services in Industry 4.0 Supply Networks.	IFAC-PapersOnLine	2015	Conference	Horizontal and vertical integration
Sousa et al.	Job shop schedules analysis in the context of industry 4.0	2017 International Conference on Engineering, Technology and Innovation (ICE/ITMC)	2017	Conference	Combination of majors
Tang et al.	RFID Network Planning for Wireless Manufacturing Considering the Detection Uncertainty.	IFAC-PapersOnLine	2015	Conference	CPS
Tantik and Anderl	Industrie 4.0: Using Cyber-Physical Systems for Value-Stream Based Production Evaluation.	Procedia CIRP	2016	Conference	CPS
Rüßmann et al.	Industry 4.0: The Future of Productivity and Growth in Manufacturing Industries		2015	Report	Combination of majors
Trstenjak and Cosic	Process Planning in Industry 4.0 Environment.	Procedia Manufacturing	2017	Conference	IoT&S
Upasani et al.	Distributed Maintenance Planning in Manufacturing Industries.	Computers & Industrial Engineering	2017	Journal	CPS
Wan et al.	Reconfigurable Smart Factory for Drug Packing in Healthcare Industry 4.0	IEEE Transactions on Industrial Informatics	2019	Journal	Combination of majors
Waschneck et al.	Production Scheduling in Complex Job Shops from an Industrie 4.0 Perspective: A Review and Challenges in the Semiconductor Industry.	Proc. of SamI40 workshop at i-KNOW '16	2016	Workshop	Combination of majors
Wu et al.	The Internet of Things Enabled Shop Floor Scheduling and Process Control Method Based on Petri Nets	IEEE Access	2019	Journal	IoT&S
Yang et al.	Simulation-based dynamic shop floor scheduling for a flexible manufacturing system in the industry 4.0 environment	2017 Winter Simulation Conference (WSC)	2017	Conference	Horizontal and vertical integration
Zhang et al.	Review of Job Shop Scheduling Research and Its New Perspectives under Industry 4.0.	Journal of Intelligent Manufacturing	2017	Journal	Combination of majors
Zhang et al.	The 'Internet of Things' enabled real-time scheduling for remanufacturing of automobile engines	Journal of Cleaner Production	2018	Journal	Combination of majors
Zhong et al.	Intelligent Manufacturing in the Context of Industry 4.0: A Review.	Engineering	2017	Journal	Combination of majors
Zhou et al.	A resource allocation oriented modeling method in Cyber-Physical System	2016 2nd IEEE International Conference on Computer and Communications (ICCC)	2016	Conference	CPS

Appendix 2. Second-level literature review publications

Authors	Title	Published	Year	Type	Main addressed CSA
Adeyeri et al.	Integration of agent technology into manufacturing enterprise: A review and platform for industry 4.0	2015 International Conference on Industrial Engineering and Operations Management	2015	Conference	CSA1-2
Homberger and Fink	Generic negotiation mechanisms with side payments – Design, analysis and application for decentralized resource-constrained multi-project scheduling problems.	European Journal of Operational Research	2017	Journal	CSA1-2
Hsu et al.	Real-Time near-optimal scheduling with rolling horizon for automatic manufacturing cell	IEEE Access	2016	Journal	CSA1-2
Lang et al.	Design of automated negotiation mechanisms for decentralized heterogeneous machine scheduling.	European Journal of Operational Research	2016	Journal	CSA1-2
Leitão	Agent-based distributed manufacturing control: A state-of-the-art survey.	Engineering Applications of Artificial Intelligence	2009	Journal	CSA1-2
Pinedo	Scheduling: theory, algorithms, and systems.	Springer	2012	Book	CSA1-2
Shen et al.	Applications of agent-based systems in intelligent manufacturing: An updated review.	Advanced Engineering Informatics	2006	Journal	CSA1-2
Wang et al.	Towards smart factory for industry 4.0: a self-organized multi-agent system with big data base d feedback and coordination.	Computer Networks	2016a	Journal	CSA1-2
Wang et al.	Implementing Smart Factory of Industrie 4.0: An Outlook.	International Journal of Distributed Sensor Networks	2016b	Journal	CSA1-2
Baldea and Harjunkoski	Integrated production scheduling and process control: A systematic review.	Computers & Chemical Engineering	2014	Journal	CSA3-5
Chen et al.	Process optimization with consideration of uncertainties-An overview.	Chinese Journal of Chemical Engineering	2017b	Journal	CSA3-5
Kalra and Singh	A review of metaheuristic scheduling techniques in cloud computing.	Egyptian Informatics Journal	2015	Journal	CSA3-5
Lin et al.	Smart manufacturing scheduling with edge computing using multiclass deep Q network	IEEE Transactions on Industrial Informatics	2019	Journal	CSA3-5
Monch et al.	Production planning and control for semiconductor wafer fabrication facilities: Modeling, analysis, and systems, volume vol. 52 of Operations research/computer science interfaces series	Springer	2013	Book	CSA3-5
Mouelhi-Chibani and Pierreval	Training a neural network to select dispatching rules in real time.	Computers & Industrial Engineering	2010	Journal	CSA3-5
Nguyen	Genetic programming for production scheduling: a survey with a unified framework.	Complex & Intelligent Systems	2017	Journal	CSA3-5
Olafsson and Li	Learning effective new single machine dispatching rules from optimal scheduling data.	International Journal of Production Economics	2010	Journal	CSA3-5
Ota et al.	ORACLE: Mobility control in wireless sensor and actor networks	Computer Communications	2012	Journal	CSA3-5
Priore et al.	A review of machine learning in dynamic scheduling of flexible manufacturing systems.	Artificial Intelligence for Engineering Design, Analysis and Manufacturing archive	2001	Journal	CSA3-5

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Authors	Title	Published	Year	Type	Main addressed CSA
Shahzad and Mebarki	Learning Dispatching Rules for Scheduling: A Synergistic View Comprising Decision Trees, Tabu Search and Simulation.	Computers	2016	Journal	CSA3-5
Shiue et al.	Real-time scheduling for a smart factory using a reinforcement learning approach.	Computers & Industrial Engineering	2018	Journal	CSA3-5
Song and Eldin	Adaptive real-time tracking and simulation of heavy construction operations for look-ahead scheduling.	Automation in Construction	2012	Journal	CSA3-5
Xhafa and Abraham	Metaheuristics for Scheduling in Industrial and Manufacturing Applications.	Springer	2008	Book	CSA3-5
Berenstein and Edan	Human-robot cooperative precision spraying: Collaboration levels and optimization function.	IFAC Proceedings Volumes	2012	Conference	CSA6-7
Bochmann et al.	Human-robot Collaboration in Decentralized Manufacturing Systems: An Approach for Simulation-based Evaluation of Future Intelligent Production.	Procedia CIRP	2017	Conference	CSA6-7
Bogner et al.	Optimised scheduling in human–robot collaboration – a use case in the assembly of printed circuit boards.	International Journal of Production Research	2018	Journal	CSA6-7
Chen et al.	Optimal Subtask Allocation for Human and Robot Collaboration Within Hybrid Assembly System.	IEEE Transactions on Automation Science and Engineering	2014	Conference	CSA6-7
Gombolay et al.	Human-Machine Collaborative Optimization via Apprenticeship Scheduling.	Journal of Artificial Intelligence Research	2018	Journal	CSA6-7
Graessler and Poehler	Integration of a digital twin as human representation in a scheduling procedure of a cyber-physical production system	2017 IEEE International Conference on Industrial Engineering and Engineering Management (IEEM)	2017	Conference	CSA6-7
Lino Ferreira et al.	A pilot for proactive maintenance in industry 4.0.	2017 IEEE 13th International Workshop on Factory Communication Systems	2017	Workshop	CSA6-7
Mohan and Bhat	Joint Goal Human Robot collaboration-From Remembering to Inferring.	Procedia Computer Science	2018	Conference	CSA6-7
Nikolakis et al.	Dynamic scheduling of shared human-robot manufacturing operations.	Procedia CIRP	2018	Conference	CSA6-7
Ore et al.	Human – Industrial Robot Collaboration: Application of Simulation Software for Workstation Optimisation.	Procedia CIRP	2016	Conference	CSA6-7
Tsarouchi et al.	On a human–robot workplace design and task allocation system.	International Journal of Computer Integrated Manufacturing	2017a	Journal	CSA6-7
Tsarouchi et al.	On a human-robot collaboration in an assembly cell.	International Journal of Computer Integrated Manufacturing	2017b	Journal	CSA6-7
Zanchettin et al.	Prediction of human activity patterns for human-robot collaborative assembly tasks	IEEE Transactions on Industrial Informatics	2018	Journal	CSA6-7